AI Agent Planning Document – Hotel Management System

Project Overview

Website Topic and Purpose

This project is an **AI-powered hotel management platform** designed to streamline hotel operations and improve hotel performance. The system analyzes factors such as occupancy rates, room information, weather conditions, seasonality, and local events to provide **dynamic pricing recommendations**. Additionally, it automatically analyzes customer reviews and generates **tasks for cleaning or maintenance** when necessary, enhancing customer satisfaction and operational efficiency.

Unlike traditional booking websites, this platform focuses on **hotel management and decision support**, rather than only reservations.

Target Users

- **Hotel Managers:** Track room occupancy, revenue, and AI recommendations; approve and manage tasks.
- **Hotel Staff:** Track and complete cleaning or maintenance tasks assigned by the system.

Core Features

- **User and Role Management:** Different access rights for managers and staff.
- **Room and Hotel Management:** Manage room details, pricing, capacity, and occupancy.
- **Customer Review Analysis:** Detect cleaning, maintenance, and other operational issues from reviews.
- **AI-Powered Dynamic Pricing:** Recommend optimal pricing based on occupancy, season, events, and competitors.
- **Automated Task Generation:** Automatically create operational tasks (cleaning/maintenance) based on AI analysis.
- **Performance and Revenue Analytics:** Reports on hotel performance, occupancy, revenue, and customer satisfaction.

AI Agent Concept

Problems the AI Agent Solves

- Hotels receive a high volume of customer reviews; manual analysis is time-consuming and can miss critical issues.
- Revenue optimization, occupancy, and seasonal pricing are difficult to manage manually.
- Delayed operational actions can reduce customer satisfaction.

The AI agent will **analyze reviews, generate tasks, and provide dynamic pricing recommendations**, enabling faster and more effective operational management.

AI Agent Types

1. **Review Analysis Agent (Evaluator):**
    - Analyzes customer reviews.
    - Detects cleaning, maintenance, noise, or service issues.
    - Generates tasks for the manager panel (cleaning/maintenance).
2. **Task Generation Agent (Automated Task Agent):**
    - Uses output from the Review Analysis Agent to create tasks.
    - Assigns tasks to staff panels for operational execution.
3. **Dynamic Pricing Agent:**
    - Analyzes hotel occupancy, season, local events, and competitor pricing.
    - Suggests optimal room rates to maximize revenue.
    - Final pricing decision remains with the manager.
4. **Management Assistant Agent (Assistant)**

    - Generating performance reports
    - Providing operational recommendations
    - Summarizing hotel status

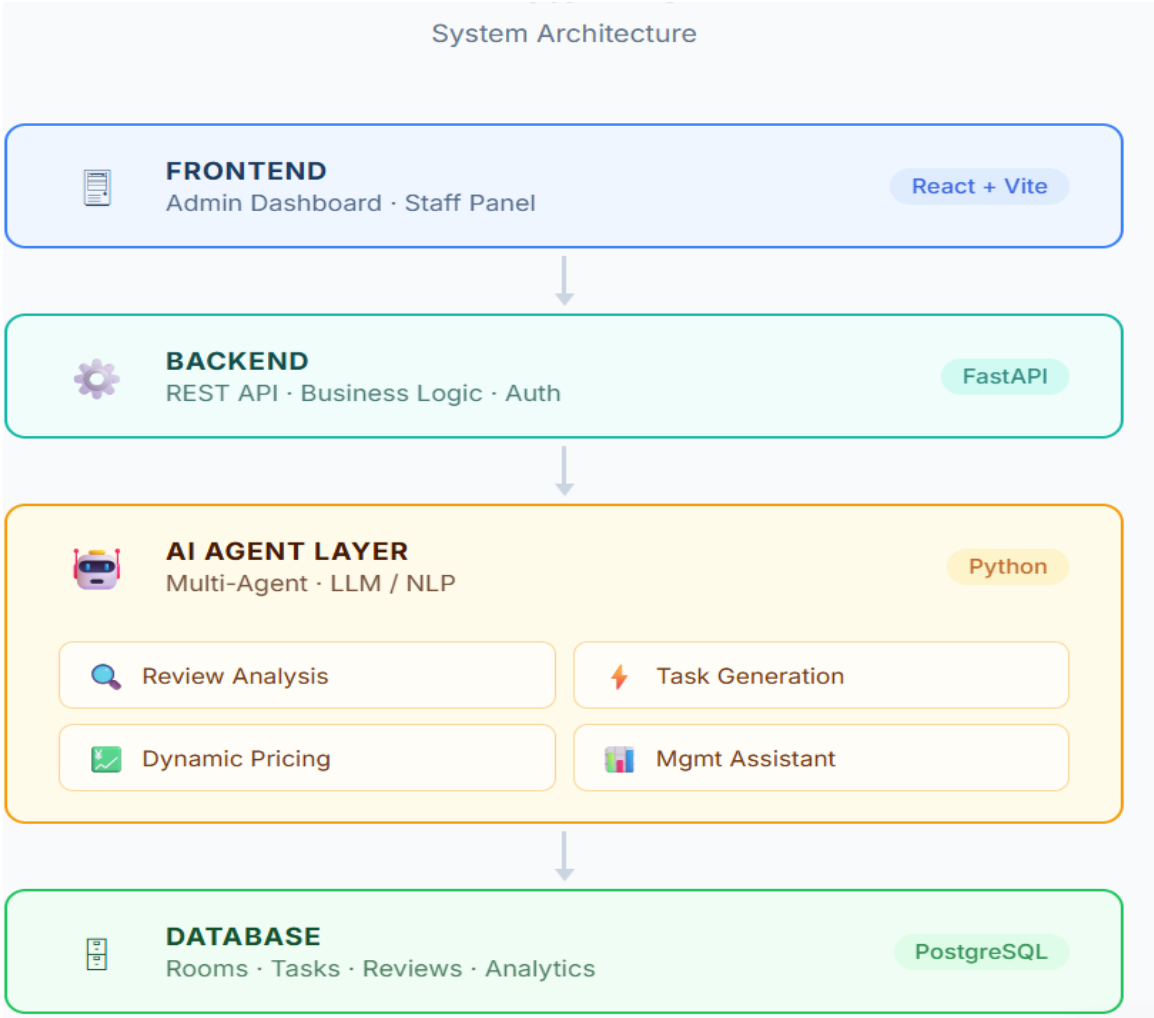
User Interaction

- Users primarily interact through the **manager dashboard**.
- AI-generated analysis, recommendations, and tasks are automatically displayed.
- Managers can generate AI reports about hotel performance.
- Future implementation may include a **chat-based AI assistant** for queries such as:
    - “Which rooms require cleaning or maintenance this week?”
    - “Should we increase prices for high-demand rooms next week?”

System Architecture (High-Level)

The platform is organized into five distinct layers with clearly defined responsibilities and interfaces. Data flows vertically from user interaction at the frontend, through the backend and AI agent layer, down to the PostgreSQL database — with external APIs feeding the AI agents for enriched decision-making.

FRONTEND *React + Vite + TypeScript*	Manager Dashboard · Staff Panel	AI Recommendation Display	Dynamic Pricing UI · Reports	JWT Role-Based Route Guards
BACKEND *FastAPI · Python 3.12*	REST API Endpoints · Business Logic	Auth (OAuth2 + JWT)	AI Agent Orchestration	SQLAlchemy ORM · Alembic
AI AGENTS *Multi-Agent · LLM / NLP*	Review Analysis Agent · Task Generation Agent	Dynamic Pricing Agent	Management Assistant Agent	Event-Driven + Scheduled Triggers
DATABASE *PostgreSQL 15 · Docker*	users · rooms · room_pricing	housekeeping_tasks · maintenance_tickets	guest_reviews · ai_analysis_results	pricing_recommendations · agent_logs
EXTERNAL *Third-Party APIs*	Weather API · Local Events API	Competitor Rate Feed	LLM / NLP Cloud Service	Booking Platform Review Feed



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